

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Second Semester M.Tech(CBS)(CBCS) Degree Examinations

May – June 2025

22CBS221 – SECURITY AND RESILIENCE

Duration: 3 Hours

Max. Marks:100

Note: Answer Five full questions choosing Two full questions from Unit – I & Unit – II each and One full question from Unit – III.

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain the need for reformulating the concept of command and control.	5	L*2	1	1
	b) Explain the different types of clouds based on their visibility with a neat diagram.	10	L2	2	1
	c) Explain the future Command and Control (C2) systems with a neat diagram.	5	L2	1	1
2.	a) Why is important to apply cloud technologies to build data centers for defense and security? Explain the benefits of applying cloud technologies for defense and security.	10	L2	2	1
	b) Explain SaaS, PaaS, and IaaS cloud services with suitable examples.	10	L2	2	1
3.	a) What is cloud computing? Explain its benefits.	10	L2	2	1
	b) What is cyber resilience? Explain its importance.	5	L2	1	1
	c) Explain the requirements of mobile data centers.	5	L2	2	1
Unit – II					
4.	a) Explain the situational awareness model developed by Endsley. Explain the framework for the situational awareness of the critical infrastructure.	10	L2	3	1
	b) What is a smart grid? Explain the network architecture of a smart grid with a neat diagram.	10	L2	3	1
5.	a) Explain the requirements for the cyber security in the smart grids.	10	L2	3	1
	b) What is Controller Area Network? Explain CAN message and frames with suitable diagrams.	10	L2	4	1
6.	a) Explain the Generalized Net (GN) model with a neat diagram.	10	L2	4	1
	b) Describe self-discrepancy framework with a neat diagram.	10	L1	4	1
Unit – III					
7.	a) Explain the various causes for the rise in internal threats to critical infrastructure. Explain the different types of internal threats affecting the critical infrastructure.	10	L2	5	1
	b) Explain the different ways of protecting the critical infrastructure from internal threats.	10	L2	5	1
8.	a) Explain the basic concept of cyber range and cyber range capabilities.	10	L2	5	1
	b) Explain how AR/VR technologies help in the learning process.	10	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
